

INTERNATIONAL YOUTH ENVIRONMENTAL CHALLENGE 2026 · SENIOR CATEGORY

RootStake: Giving Every Plant in Singapore a Voice

Team RootedSG · Singapore October 2026 857 words

3,043	51.2%	4 / 4	14
Singapore residents surveyed	want plants they do not have	plants healthy after a 15-day test	probe designs built and refined

Across Singapore's corridors and balconies, plants die in the same quiet way. The soil dries too fast in an afternoon heatwave, or stays waterlogged after a careful owner waters "just in case". By the time a leaf curls, the damage below the surface is done. The plant had no way to say what it needed, and its owner had no way to see it. RootStake exists to end that silence.

The invisible failure

Singapore imports over 90% of its food. Singapore Food Story 2 aims to grow 20% of the vegetables the country eats locally by 2035, up from about 8% in 2024. Farms will carry most of that goal, but there is a second, uncounted growing space: the corridors, balconies and windowsills of more than a million public homes.

Our survey of 3,043 residents shows how much of it is lost. Interest is high: 70.9% rated their interest in growing at home 4 or 5 out of 5. Yet 51.2% have stopped growing or never started. The reasons are not space or money. Forgetting or lack of time (36.1%), not knowing who to ask (26.7%) and losing motivation (24.0%) dominate, and 16.3% do not know where to look for help. The real problem is information: nobody can see what is happening under the soil until it is too late.

A nervous system for a single plant

RootStake gives a plant the senses, reflexes and voice it lacks, in a closed loop of four parts we designed and 3D-printed ourselves.

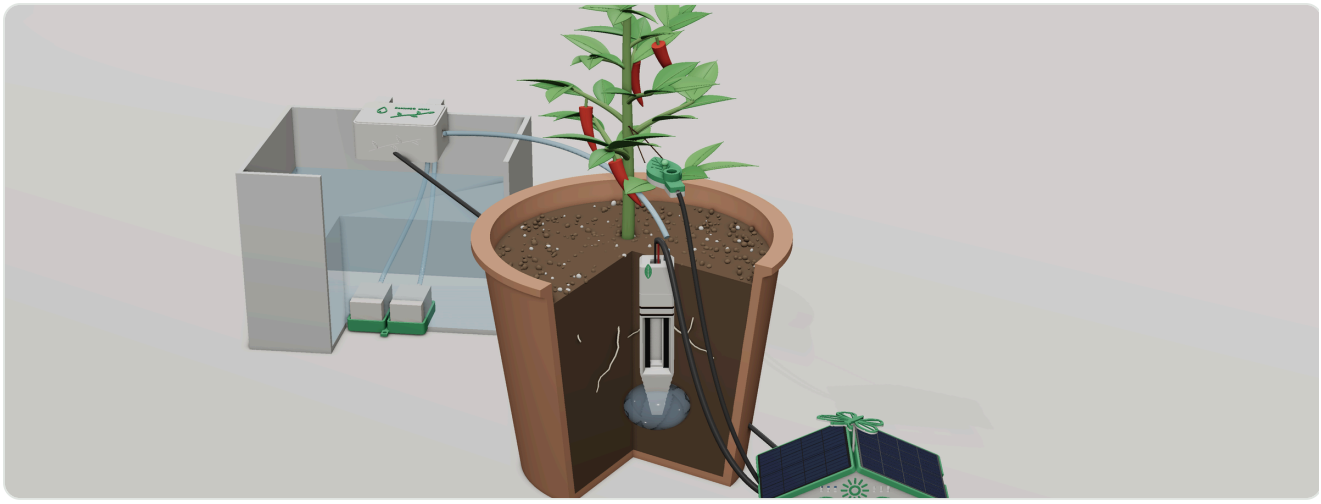


Figure 1. The complete RootStake loop: probe in the root zone, leaf electrode patch, solar module and pump station (render of our CAD).

The **probe** sits in the root zone. A moisture fork measures water where roots actually drink, dedicated sensors track nitrogen, phosphorus and magnesium, and copper conductivity bands wrap its buried shoulder. Hidden inside, a fertigation channel curves to the centre of a chisel-shaped tip, releasing water and liquid fertiliser directly among the roots.

A **leaf electrode patch** reads the plant's electrical signals linked to stress, while a **DHT11**, shaded beneath a vented roof, measures the air the leaves are breathing.

The **module** is the brain: an ESP32-S3 microcontroller powered by two 70 × 70 mm solar panels, needing no socket. The **pump station** is the muscle: two 5 V pumps, one for water and one for fertiliser, merged through a Y-connector into a single tube.

Intelligence at the edge

RootStake does not water on a timer. Every reading is calibrated against dry and wet references. The module tracks how fast the soil is drying, using the median of hourly changes so one noisy reading cannot trigger a false alarm, and forecasts when the plant will next need water. It accounts for vapour pressure deficit, so a hot, dry afternoon brings the next watering forward. A hysteresis rule starts the pump below the plant's lower moisture limit and stops it at the upper limit, preventing both drought and overwatering.

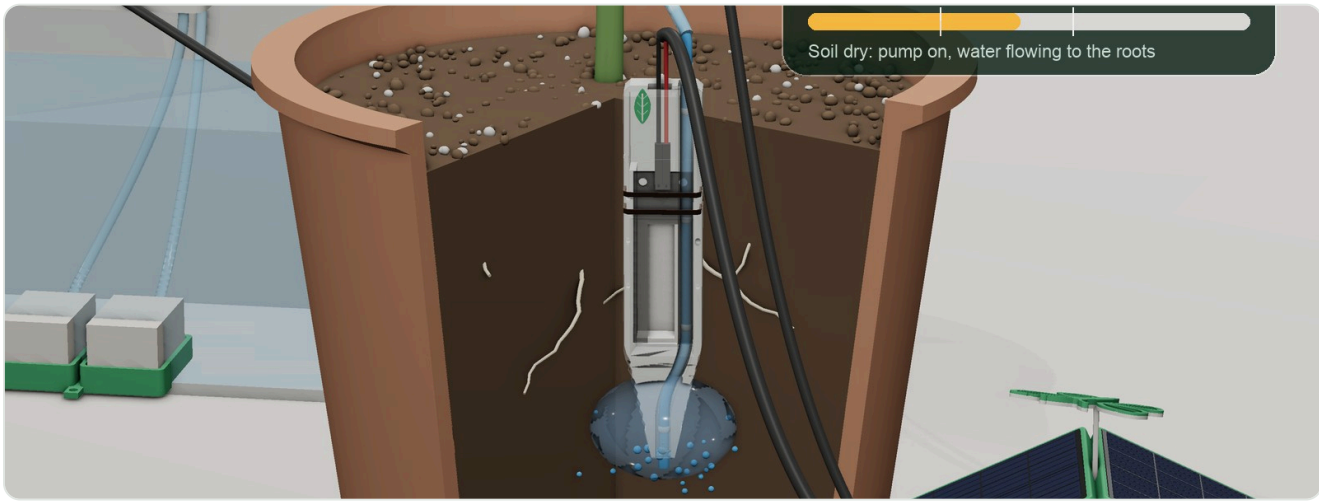


Figure 2. Water travels down the hidden channel and out of the probe tip, wetting the root zone until the upper moisture limit is reached.

Each reading travels over Bluetooth Low Energy in a compact, checksum-verified packet to the ROOTED SG app on iOS and Android, where Mojo, the app's companion, turns data into one plain instruction. Even privacy is engineered: community insights appear only when at least three probes from two households contribute, so no home can be identified.

Built, broken and rebuilt

RootStake is a working prototype, not a concept. The probe alone went through 14 designs. Version 7 printed with plastic filling its own water channel; version 8 replaced it with an open channel the tube lays into. Later versions moved the outlet to the centre, added a chisel point and saw teeth for penetrating soil, and deepened the channel. Every failure became a design rule.

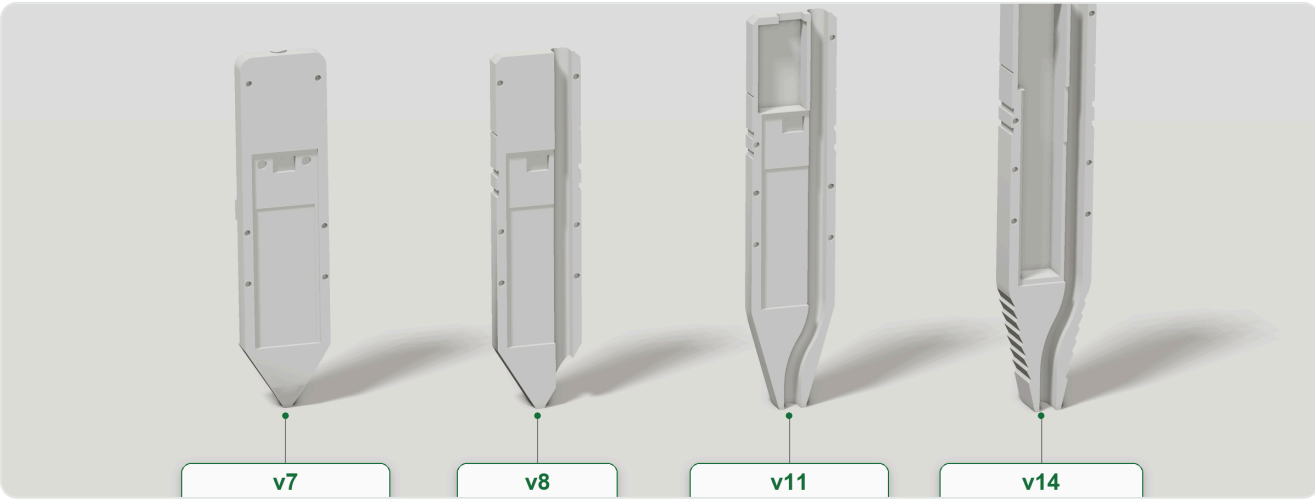


Figure 3. Four of the 14 probe designs. Each failure became a design rule.

We then ran RootStake for 15 days on four plants, with the system controlling all watering and feeding. All four stayed healthy. That proves the loop works; our next trial will prove how much better it is. At least 30 plants, half hand-watered as controls, will be measured for survival, water use and fertiliser use, and every nutrient reading will be compared with a laboratory soil test against a target of at least 80% agreement.

From one balcony to a distributed farm

RootStake's environmental value compounds. It waters only when the soil is dry, below the surface where less evaporates. It feeds in small doses at the roots, so less fertiliser washes into drains. And it keeps plants alive, so fewer seedlings, pots and bags of soil are thrown away and more people keep growing food.

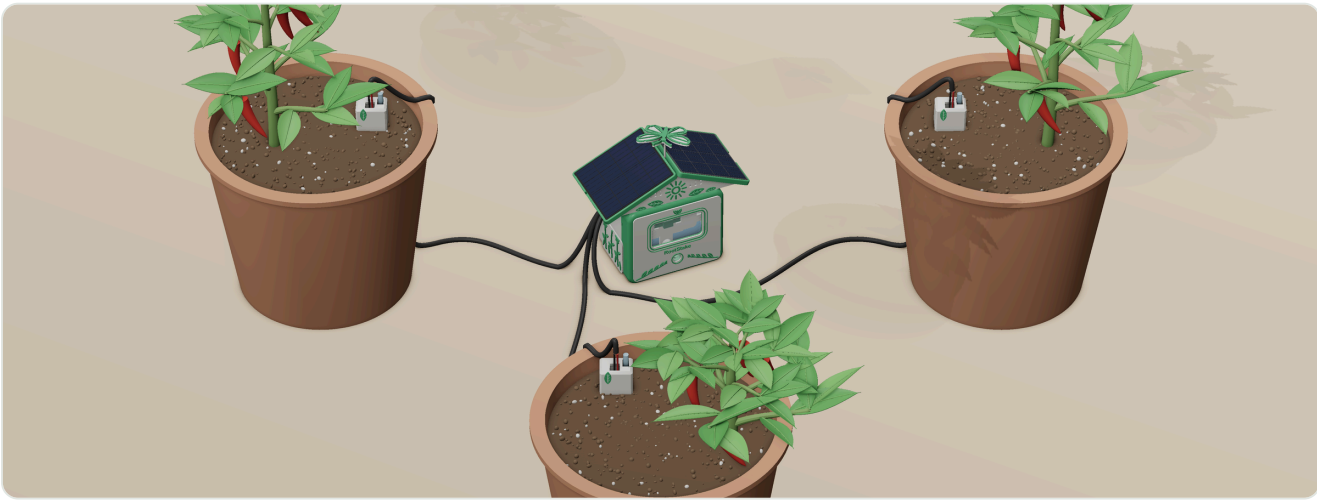


Figure 4. One module serving several plants: the cost per plant falls as a garden grows.

Because one module drives many probes, the cost per plant falls as a garden grows. The same architecture scales from a balcony to school gardens, community plots and, eventually, an anonymous city-wide map of what grows best where. If our survey reflects Singapore, roughly half a million homes want to grow and cannot. Each one that succeeds becomes a small farm on space that already exists.

The next step

Singapore's food resilience will be built mostly on farms, but it will also depend on whether ordinary households can succeed with the space they already have. RootStake turns every pot into a plant that can sense, decide and ask for help, and turns half a million balconies into a network that grows food.

Sources. Singapore Food Agency: Singapore imports over 90% of its food. Ministry of Sustainability and the Environment, oral reply on revised local production targets (Singapore Food Story 2): 20% of fibre and 30% of protein produced locally by 2035, from about 8% and 26% in 2024. RootedSG resident survey, n = 3,043 responses. RootStake field test: 15 days, 4 plants. Images: renders of the team's own CAD models and screens from the ROOTED SG app.